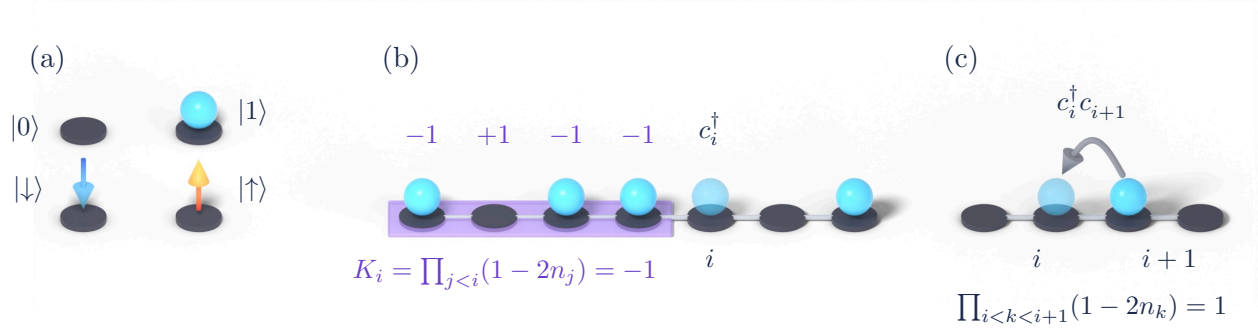

(a)
$|0\rangle$
$|1\rangle$
$|\downarrow\rangle$
$|\uparrow\rangle$
(b)
-1
$+1$
-1
-1
$c_i^\dagger$
$K_i = \prod_{j<i}(1-2n_j) = -1$
i
(c)
$c_i^\dagger c_{i+1}$
i
$i+1$
$\prod_{i<k<i+1}(1-2n_k) = 1$